

NUEIP: Delivering a highly available company management software with an auto- scalable cloud infrastructure

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About NUEIP

NUEIP leverages
scalable infrastru
preventive securi
of Google Cloud t
highly available a
company manag

Founded in 2015, NUEIP is a Taiwanese cloud enterprise software provider dedicated to delivering easy-to-use digital solutions that simplify company management. With its platforms for human resource management, customer relationship management and business intelligence, it currently provides services to more than 2,000 enterprise customers and 100,000+ users.

Industries: Professional Services

solutions while re
operational and
maintenance cos

Location: Taiwan

Google Cloud results

- Shortens content delivery time by up to 10X with Cloud CDN
- Prevents disruptions caused by DDoS attacks via Google Cloud Armor and VPC
- Facilitates computing environment duplication for security tests with Compute Engine

Zero c
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years
deploy

Cloud Storage

Google Cloud (<https://cloud.google.com/>) Compute Engine (<https://cloud.google.com/compute/>),

Cloud CDN (<https://cloud.google.com/cdn/>)

(<https://cloud.google.com/armor/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Load Balancing

(<https://cloud.google.com/load-balancing/>)

Cloud Storage (<https://cloud.google.com/storage/>)

(<https://cloud.google.com/vpc/>)

Compute Engine

(<https://cloud.google.com/compute/>)

Google Cloud Armor

(<https://cloud.google.com/armor/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Cloud Memorystore

(<https://cloud.google.com/memorystore/>)

Virtual Private Cloud

(<https://cloud.google.com/vpc/>)

- Reduces
10X
operational
costs
through
cost-
efficient
computing
resource
utilization
enabled by
GKE

As more companies in Taiwan embark on digital transformation, the demand for high-performance company management softwares is increasing. Company executives are constantly searching for solutions that can help simplify and streamline business operations.

NUEIP (<https://www.nueip.com/en/aboutus>) is dedicated to bringing easy-to-use, top-notch digital solutions for company management by leveraging the most advanced technologies. Founded in 2015, the Taiwanese software provider is currently offering four business management solutions with innovative features to more than 2,000 enterprise customers. One of its products is NUEIP's human resource management platform, which allows employees to clock in and out on their mobile devices and have shift schedules synchronized in real time on their digital work calendars.

NUEIP first deployed its services on a private cloud. It needed to do manual backups, was unable to filter out suspicious connections and had limited computing capabilities. The company therefore needed to invest more time and effort to guarantee the availability and security of its services. When a traffic spike happened, its systems could only maintain 30% of the normal connectivity, because adding extra computing resources required a total shutdown of its platforms.

"The traffic on our human resource management system can instantly rise by five times, because company employees tend to clock in and out at around the same time," explains Mars Hung, technical manager at NUEIP. "On the private cloud that we previously used, we needed to shut down the system to deploy more computing resources for higher traffic, but it was not possible to do so when our services were most needed, so we could only compromise on poor connectivity."

To improve the quality of its services, NUEIP decided to replace its cloud infrastructure with a cloud platform that provides more resources and better technologies. In July 2018, it migrated its services to Google Cloud (<https://cloud.google.com/>) because of its data center in Taiwan and the intuitive nature of its cloud tools.

"The fact that Google Cloud has a data center in Taiwan is a great advantage for us, because it provides greater information security and doubles the connection speed for our customers in Taiwan," says Hung. "Besides, the abundant out-of-the-box

features of Google Cloud enable our team to quickly set up a new cloud infrastructure."

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Ensuring high availability through autoscaling and load balancing

According to Hung, the configurations on Google Cloud are so intuitive that it requires only little prior knowledge for the NUEIP engineers to get the hang of the new cloud tools. As a result, the team was able to build a well-functioning cloud computing environment in 10 days, which is four times shorter than estimated. The Google Cloud team also helped reduce the development work by giving suggestions to improve NUEIP's virtual machine (VM) server structure.

NUEIP now runs its platforms for human resource management, customer relationship management and business intelligence with the VMs on Compute Engine (<https://cloud.google.com/compute>) and the database on Cloud SQL (<https://cloud.google.com/sql>). The autoscaling feature of Compute Engine ensures that NUEIP always has sufficient computing resources to handle any traffic spike, while Cloud SQL gives more computing capacity by separating read and write operations. On top of that, Memorystore (<https://cloud.google.com/memorystore>) can cache incoming data for processing later when

the flow is high. This way, the NUEIP is capable of swiftly tackling sudden flow surges.

"With Compute Engine and Cloud SQL, we no longer need to worry about not having enough computing resources to run our services" notes Hung. "Since our adoption of Google Cloud, our company management platforms haven't encountered any delay or downtime due to high traffic, which has greatly enhanced the user experience."

To further increase the performance of its systems, NUEIP employs Cloud Load Balancing (<https://cloud.google.com/load-balancing>) to distribute traffic across different servers for better load management, and Cloud CDN (<https://cloud.google.com/cdn>) to cache the static content on its systems for faster content delivery. By caching content in servers that are closer to users, Cloud CDN has helped shorten the response time for access requests from 100 to 10-50 milliseconds.

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Maintaining a high level of information security

Since NUEIP's company management systems process a large amount of business data, guaranteeing information security is one of its top priorities. Besides storing its customer data in [Cloud](#)

Storage (<https://cloud.google.com/storage>) hosted in Taiwan's data center, NUEIP also uses the encrypted connection of Cloud Load Balancing to ensure that data transferring to and from its systems is always secure.

Before migrating to Google Cloud, NUEIP suffered from two DDoS attacks that led to system overload and subsequent crash, because its previous cloud environment could neither detect threats nor immediately scale up computing resources. To better prevent cyber attacks, the company now leverages Google Cloud Armor (<https://cloud.google.com/armor>) that sends instant alerts when potential attacks are detected and blocks malicious traffic, and has set up a firewall on Virtual Private Cloud (<https://cloud.google.com/vpc>) (VPC) that can deny suspicious connections.

"DDoS attacks are a common threat faced by almost all the software providers," notes Hung. "Thanks to the security tools of Google Cloud like Google Cloud Armor and VPC, our services haven't been affected by any attack since the migration. The auto-scalable computing resources on Google Cloud also give us more time to clear threats by digesting any amount of flow increase and preventing our systems from crashing."

Automating VMs for better resource utilization

NUEIP regularly duplicates the computing environments of its service platforms for security tests. Because Compute Engine can automatically

create VMs with the same configuration, the NUEIP team can clone its computing architecture within one week, instead of several weeks when it deployed its systems on a private cloud.

Currently, NUEIP relies on Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>) (GKE) to run its business intelligence platform, which requires a highly flexible scheme to adjust computing resources, because the amount of data it needs to process varies greatly from day to day. The autopilot feature of GKE helps NUEIP use computing resources cost-efficiently by automatically and immediately adding or reducing VMs according to demand. As a result, the company has lowered its operational costs by 10X.

Hung says that the auto-update features of the other Google Cloud tools have also enabled NUEIP to reduce its maintenance expenses. "The high level of automation on Google Cloud has helped us free more resources from maintenance for product development. Overall, we have reduced our labor demand for system maintenance by 70%," he adds.

Supporting product optimization and business expansion

Moving forward, NUEIP will continue optimizing and expanding its software products, and is set to enter the markets outside Taiwan in the near future. The company is currently developing its new product with Cloud Spanner (<https://cloud.google.com/spanner>) that supports a more reliable and scalable global database, and plans to use Pub/Sub

(<https://cloud.google.com/pubsub>) and Cloud Tasks (<https://cloud.google.com/tasks>) to realize more flexible data management. Richard Chang, vice president at NUEIP says that the global network and fully-managed services of Google Cloud can greatly facilitate the company's business expansion abroad, because it doesn't have to worry too much about infrastructure deployment and maintenance.

"Over the past three years, Google Cloud has helped us realize more than 10X business growth by supporting our highly available and secure business management solutions. Seeing this, we're convinced that we can always rely on Google Cloud for continuous product optimization and market expansion," adds Chang.

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